Quiz Makeup

Q1. A university database schema is provided below:

Student(***student_id***, name, major, year) -- PK: student_id
-- FK: major references Departments(dept_name)

Course***(course_id***, title, credits) -- PK: course_id

Enrollment(***student_id***, ***course_id***, grade) -- PK: student_id, course_id
-- FK: student_id references Student(student_id)
-- FK: course_id references Course(course_id)

Department(***dept_name***, building, budget) -- PK: dept_name

Write Relational Algebra (RA) for the following:

a. [2 Marks] List the names and majors of all students in their final year (year = 4).
b. [2 Marks] List the names of students and the titles of courses they are enrolled in, where the course has more than 3 credits.
c. [3 Marks] List the names of students, the titles of courses they are enrolled in, and the building where their major's department is located, for students who are taking courses with exactly 3 credits.
d. [1 Marks] Rename the "Department" relation to "Dept".
e. [2 Marks] Convert the following sql statements into its appropriate Relational Algebra (RA) expression:

```
SELECT name, title
FROM Student, Enrollment, Course
WHERE Student.student_id = Enrollment.student_id
AND Enrollment.course_id = Course.course_id
AND Course.credits > 3;
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a. $\pi_{name, major}\ (\sigma_{year = 4}\ (Student))$
b. $R \leftarrow (Student \bowtie_{student_id = student_id} Enrollment \bowtie_{course_id = course_id} Course)$
$\pi_{name, title}\ (\sigma_{credits > 3}\ (R))$
c. $R \leftarrow (Student \bowtie_{student_id = student_id} Enrollment \bowtie_{course_id = course_id} Course)$
$T \leftarrow (R \bowtie_{major = dept_name} Department)$
$\pi_{name, title, building}\ (\sigma_{credits = 3}\ (T))$
d. $\rho_{Dept}\ (Department)$
e. $R \leftarrow (Student \bowtie_{student_id = student_id} Enrollment \bowtie_{course_id = course_id} Course)$
$\pi_{name, title}\ (\sigma_{credits > 3}\ (R))$